Table 1: L2 (TDOST classes), all classes — comparable to TDOST / DeepCASAS; $m = all$ labelled windows per cluster. W/M = weighted/macro F1.

| Method | Repr. | $K$ | Milan | | Aruba | | Cairo | | Average | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | W-F1 | M-F1 | W-F1 | M-F1 | W-F1 | M-F1 | W-F1 | M-F1 |
| k-means | Sensor hist. | 20 | 0.79 | 0.45 | 0.88 | 0.53 | 0.76 | 0.33 | 0.81 | 0.44 |
| k-means | Sensor hist. | 100 | 0.81 | 0.49 | **0.89** | 0.59 | 0.79 | 0.36 | 0.83 | 0.48 |
| k-means | Sensor+ToD hist. | 20 | 0.72 | 0.31 | 0.80 | 0.31 | 0.84 | 0.40 | 0.79 | 0.34 |
| k-means | Sensor+ToD hist. | 100 | 0.82 | 0.50 | 0.88 | 0.52 | 0.87 | **0.49** | 0.86 | **0.51** |
| k-means | MLM | 20 | 0.80 | 0.45 | **0.89** | 0.55 | 0.76 | 0.35 | 0.82 | 0.45 |
| k-means | MLM | 100 | 0.82 | 0.50 | **0.89** | 0.59 | 0.81 | 0.43 | 0.84 | **0.51** |
| k-means | MLM-time | 20 | 0.81 | 0.47 | 0.89 | 0.50 | 0.78 | 0.33 | 0.82 | 0.43 |
| k-means | MLM-time | 100 | 0.81 | 0.50 | 0.89 | 0.57 | 0.81 | 0.45 | 0.84 | **0.51** |
| k-means | MLM + ToD | 20 | 0.72 | 0.31 | 0.80 | 0.31 | 0.86 | 0.43 | 0.79 | 0.35 |
| k-means | MLM + ToD | 100 | **0.82** | 0.50 | **0.89** | 0.50 | **0.89** | 0.47 | **0.87** | 0.49 |
| k-means | MLM-time + ToD | 20 | 0.72 | 0.31 | 0.78 | 0.28 | 0.86 | 0.41 | 0.79 | 0.33 |
| k-means | MLM-time + ToD | 100 | **0.82** | **0.51** | 0.88 | 0.54 | 0.88 | 0.47 | 0.86 | **0.50** |
| SCAN | MLM | 20 | 0.77 | 0.34 | 0.85 | 0.42 | 0.79 | 0.35 | 0.80 | 0.37 |
| SCAN | MLM | 100 | 0.80 | 0.45 | **0.89** | 0.60 | 0.82 | 0.40 | 0.84 | 0.48 |
| SCAN | MLM-time | 20 | 0.75 | 0.30 | 0.88 | 0.55 | 0.77 | 0.32 | 0.80 | 0.39 |
| SCAN | MLM-time | 100 | 0.81 | 0.47 | **0.89** | **0.60** | 0.82 | 0.45 | 0.84 | **0.51** |
| DEC | MLM | 20 | 0.78 | 0.35 | 0.88 | 0.46 | 0.77 | 0.33 | 0.81 | 0.38 |
| DEC | MLM | 100 | 0.80 | 0.45 | 0.87 | 0.54 | 0.79 | 0.36 | 0.82 | 0.45 |
| DEC | MLM-time | 20 | 0.77 | 0.34 | **0.89** | 0.47 | 0.75 | 0.29 | 0.81 | 0.37 |
| DEC | MLM-time | 100 | 0.80 | 0.43 | **0.89** | 0.55 | 0.80 | 0.38 | 0.83 | 0.45 |

Table 2: L2, rare classes masked from the average; $m = all$ labelled windows per cluster. W/M = weighted/macro F1.

| Method | Repr. | $K$ | Milan | | Aruba | | Cairo | | Average | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | W-F1 | M-F1 | W-F1 | M-F1 | W-F1 | M-F1 | W-F1 | M-F1 |
| k-means | Sensor hist. | 20 | 0.80 | 0.56 | 0.88 | 0.78 | 0.76 | 0.38 | 0.81 | 0.58 |
| k-means | Sensor hist. | 100 | 0.82 | 0.61 | **0.89** | 0.80 | 0.79 | 0.41 | 0.83 | 0.61 |
| k-means | Sensor+ToD hist. | 20 | 0.72 | 0.38 | 0.81 | 0.50 | 0.84 | 0.47 | 0.79 | 0.45 |
| k-means | Sensor+ToD hist. | 100 | 0.82 | 0.61 | 0.88 | 0.77 | 0.88 | 0.52 | 0.86 | 0.63 |
| k-means | MLM | 20 | 0.80 | 0.57 | **0.89** | **0.81** | 0.77 | 0.40 | 0.82 | 0.59 |
| k-means | MLM | 100 | 0.82 | **0.62** | **0.90** | 0.80 | 0.81 | 0.44 | 0.84 | 0.62 |
| k-means | MLM-time | 20 | 0.81 | 0.59 | 0.89 | 0.80 | 0.78 | 0.39 | 0.83 | 0.59 |
| k-means | MLM-time | 100 | 0.81 | **0.62** | 0.89 | 0.80 | 0.81 | 0.44 | 0.84 | 0.62 |
| k-means | MLM + ToD | 20 | 0.73 | 0.38 | 0.80 | 0.49 | 0.86 | 0.51 | 0.80 | 0.46 |
| k-means | MLM + ToD | 100 | **0.83** | **0.62** | **0.90** | 0.80 | **0.90** | **0.55** | **0.87** | **0.66** |
| k-means | MLM-time + ToD | 20 | 0.73 | 0.39 | 0.78 | 0.45 | 0.86 | 0.48 | 0.79 | 0.44 |
| k-means | MLM-time + ToD | 100 | **0.83** | **0.62** | 0.88 | 0.79 | 0.89 | 0.53 | 0.87 | 0.65 |
| SCAN | MLM | 20 | 0.78 | 0.42 | 0.85 | 0.58 | 0.79 | 0.41 | 0.81 | 0.47 |
| SCAN | MLM | 100 | 0.81 | 0.57 | **0.90** | 0.79 | 0.82 | 0.46 | 0.84 | 0.61 |
| SCAN | MLM-time | 20 | 0.76 | 0.37 | 0.88 | 0.78 | 0.78 | 0.38 | 0.81 | 0.51 |
| SCAN | MLM-time | 100 | 0.81 | 0.58 | **0.89** | 0.79 | 0.82 | 0.44 | 0.84 | 0.61 |
| DEC | MLM | 20 | 0.79 | 0.43 | 0.89 | 0.74 | 0.78 | 0.38 | 0.82 | 0.52 |
| DEC | MLM | 100 | 0.80 | 0.56 | 0.87 | 0.74 | 0.80 | 0.42 | 0.83 | 0.57 |
| DEC | MLM-time | 20 | 0.78 | 0.43 | **0.89** | 0.75 | 0.76 | 0.34 | 0.81 | 0.51 |
| DEC | MLM-time | 100 | 0.80 | 0.54 | **0.89** | 0.76 | 0.81 | 0.44 | 0.83 | 0.58 |

Table 3: L1 (raw CASAS activities), rare classes masked from the average; $m = all$ labelled windows per cluster. W/M = weighted/macro F1.

| Method | Repr. | $K$ | Milan | | Aruba | | Cairo | | Average | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | W-F1 | M-F1 | W-F1 | M-F1 | W-F1 | M-F1 | W-F1 | M-F1 |
| k-means | Sensor hist. | 20 | 0.79 | 0.58 | 0.87 | 0.60 | 0.29 | 0.26 | 0.65 | 0.48 |
| k-means | Sensor hist. | 100 | 0.81 | 0.62 | **0.89** | 0.65 | 0.32 | 0.30 | 0.67 | 0.52 |
| k-means | Sensor+ToD hist. | 20 | 0.70 | 0.38 | 0.80 | 0.38 | 0.72 | 0.44 | 0.74 | 0.40 |
| k-means | Sensor+ToD hist. | 100 | 0.81 | 0.63 | 0.87 | 0.63 | 0.76 | **0.57** | 0.82 | 0.61 |
| k-means | MLM | 20 | 0.80 | 0.59 | **0.89** | 0.65 | 0.30 | 0.26 | 0.66 | 0.50 |
| k-means | MLM | 100 | 0.81 | **0.63** | **0.89** | 0.65 | 0.33 | 0.32 | 0.68 | 0.53 |
| k-means | MLM-time | 20 | 0.80 | 0.59 | 0.88 | 0.65 | 0.30 | 0.27 | 0.66 | 0.50 |
| k-means | MLM-time | 100 | 0.81 | 0.62 | 0.88 | 0.64 | 0.34 | 0.33 | 0.68 | 0.53 |
| k-means | MLM + ToD | 20 | 0.73 | 0.37 | 0.80 | 0.37 | 0.72 | 0.45 | 0.75 | 0.40 |
| k-means | MLM + ToD | 100 | 0.82 | 0.63 | **0.89** | **0.66** | **0.78** | **0.58** | **0.83** | **0.62** |
| k-means | MLM-time + ToD | 20 | 0.72 | 0.37 | 0.78 | 0.32 | 0.74 | 0.45 | 0.74 | 0.38 |
| k-means | MLM-time + ToD | 100 | **0.82** | **0.63** | 0.88 | 0.64 | 0.77 | 0.56 | **0.83** | 0.61 |
| SCAN | MLM | 20 | 0.77 | 0.47 | 0.84 | 0.46 | 0.32 | 0.26 | 0.64 | 0.40 |
| SCAN | MLM | 100 | 0.80 | 0.59 | **0.89** | 0.64 | 0.34 | 0.33 | 0.68 | 0.52 |
| SCAN | MLM-time | 20 | 0.75 | 0.44 | 0.87 | 0.60 | 0.34 | 0.28 | 0.65 | 0.44 |
| SCAN | MLM-time | 100 | 0.81 | 0.60 | **0.89** | 0.64 | 0.35 | 0.34 | 0.68 | 0.53 |
| DEC | MLM | 20 | 0.78 | 0.51 | 0.87 | 0.58 | 0.29 | 0.25 | 0.65 | 0.45 |
| DEC | MLM | 100 | 0.80 | 0.58 | 0.87 | 0.59 | 0.30 | 0.28 | 0.66 | 0.48 |
| DEC | MLM-time | 20 | 0.77 | 0.47 | **0.89** | 0.61 | 0.26 | 0.22 | 0.64 | 0.43 |
| DEC | MLM-time | 100 | 0.79 | 0.56 | **0.89** | 0.63 | 0.32 | 0.30 | 0.67 | 0.50 |